

$$2-5: 1) x(t) = \begin{cases} \cos \omega_0 t & |t| < T \\ 0 & |t| > T \end{cases}$$

$$\text{定义: } X(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \quad \text{代入 } x(t)$$

$$X(\omega) = \int_{-T}^T \cos \omega_0 t \cdot e^{-j\omega t} dt$$

$$2) \text{欧拉展开余弦函数 } \cos \omega_0 t = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \quad \text{代入}$$

$$\therefore X(\omega) = \frac{1}{2} \int_{-T}^T (e^{j\omega_0 t} + e^{-j\omega_0 t}) e^{-j\omega t} dt$$

$$= \frac{1}{2} \left[\int_{-T}^T e^{j(\omega_0 - \omega)t} dt + \int_{-T}^T e^{-j(\omega_0 + \omega)t} dt \right]$$

$$3) \text{公式: } \int_a^b e^{j\Omega t} dt = \frac{2 \sin \frac{\Omega T}{2}}{\Omega}$$

$$\therefore \int_{-T}^T e^{j(\omega_0 - \omega)t} dt = \frac{2 \sin [(\omega_0 - \omega)T]}{\omega_0 - \omega}$$

$$\int_{-T}^T e^{-j(\omega_0 + \omega)t} dt = \frac{2 \sin [(\omega_0 + \omega)T]}{\omega_0 + \omega}$$

$$\therefore X(\omega) = \frac{1}{2} \left[\frac{2 \sin [(\omega_0 - \omega)T]}{\omega_0 - \omega} + \frac{2 \sin [(\omega_0 + \omega)T]}{\omega_0 + \omega} \right] = \frac{\sin [(\omega_0 - \omega)T]}{\omega_0 - \omega} + \frac{\sin [(\omega_0 + \omega)T]}{\omega_0 + \omega}$$

$$\text{拉普拉斯法: } X(s) = \frac{1}{2} \left[\int_{-T}^T e^{-(s - j\omega_0)t} dt + \int_{-T}^T e^{-(s + j\omega_0)t} dt \right]$$

$$X(s) = \frac{\sinh [(s - j\omega_0)T]}{s - j\omega_0} + \frac{\sinh [(s + j\omega_0)T]}{s + j\omega_0}$$

$$s = j\omega \text{ 代入: } \sinh [j(\omega - \omega_0)T] = j \sin [(\omega - \omega_0)T]$$

$$\sinh [j(\omega + \omega_0)T] = j \sin [(\omega + \omega_0)T]$$

$$\therefore X(j\omega) = \frac{j \sin [(\omega - \omega_0)T]}{j(\omega - \omega_0)} + \frac{j \sin [(\omega + \omega_0)T]}{j(\omega + \omega_0)}$$

$$= \frac{\sin [(\omega - \omega_0)T]}{\omega - \omega_0} + \frac{\sin [(\omega + \omega_0)T]}{\omega + \omega_0}$$

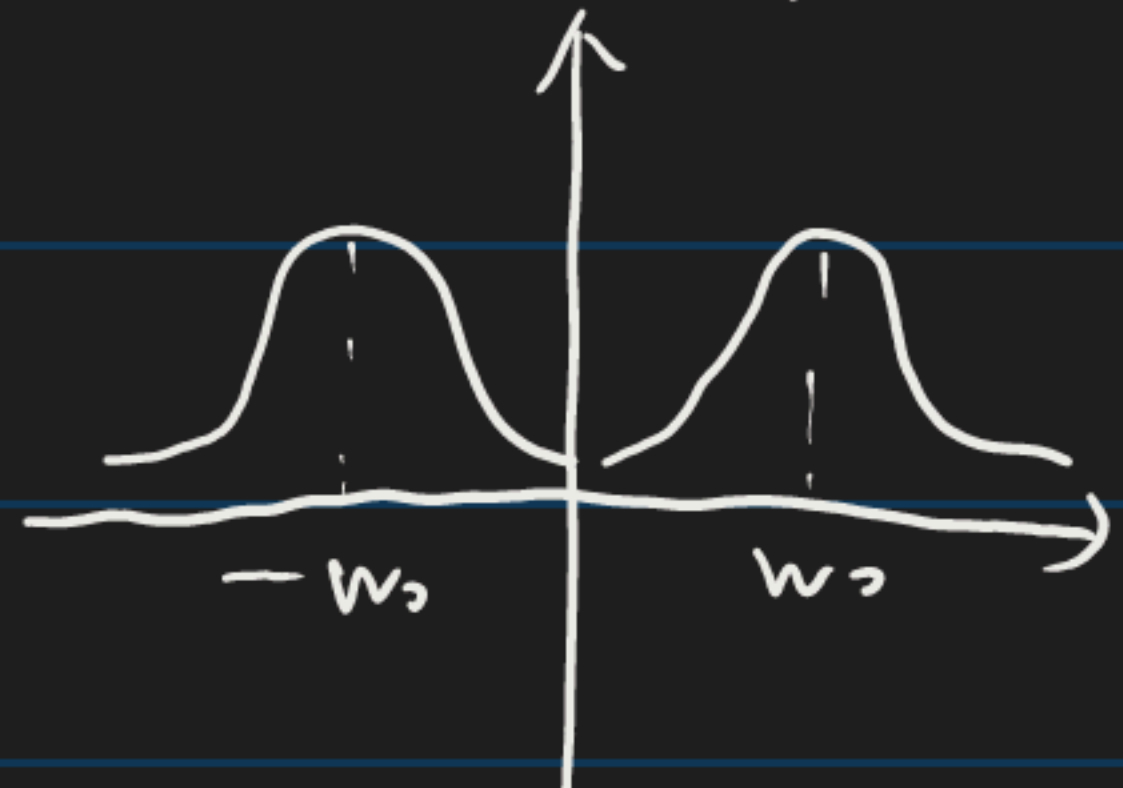
$$2-7: 1) f(t) \xrightarrow{\text{傅里叶}} F(\omega) \quad \mathcal{F}[f(t)] = F(\omega) \quad \mathcal{F}[f(t)e^{\pm j\omega_0 t}] = F(\omega \mp \omega_0)$$

$$\cos \omega_0 t = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \quad f(t) \cos \omega_0 t = \frac{1}{2} f(t) e^{j\omega_0 t} + \frac{1}{2} f(t) e^{-j\omega_0 t}$$

$$2) \mathcal{F}[f(t) \cos \omega_0 t] = \frac{1}{2} \mathcal{F}[f(t) e^{j\omega_0 t}] + \frac{1}{2} \mathcal{F}[f(t) e^{-j\omega_0 t}] = \frac{1}{2} F(\omega - \omega_0) + \frac{1}{2} F(\omega + \omega_0)$$

3) $\omega_0 > \omega_m$ 时, $F(\omega)$ 带宽为 $2\omega_m$, 调制后, 频谱 $\pm F(\omega - \omega_0)$ 搬移到 ω_0 处.

$\pm F(\omega + \omega_0)$ 搬移到 $-\omega_0$ 处. 两个边带频谱无重叠, 形成标准调幅频谱.



4) $\omega_0 < \omega_m$ 时, 搬移后, 两个频谱 $\pm F(\omega - \omega_0)$ 和 $\pm F(\omega + \omega_0)$ 在频域发生重叠.

导致频谱混叠, 无法从调幅信号中分离 $f(t)$.